Wood Provision Ecosystem Service

Spatially explicit net annual increment and overuse check: annotated R processing routine

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Introduction

This page documents the R routine that updates the wood provision ecosystem service for the European Union and makes it spatially explicit. Wood provision is the ecosystem contribution to the growth of trees, and it is measured here with the net annual increment (NAI) of the forest available for wood supply (FAWS), in thousand cubic metres over bark (Eurostat, 2026a; La Notte et al., 2021). Each step is written so that users can read it, rerun it from the repository alone and reuse the parts that generalise.

The routine has five components, in this order:

1. the **statistical component**, which builds a complete country-by-year NAI series from Eurostat `for_vol_efa` (Eurostat, 2026b) for 2000 to 2024, and reports the years 2000, 2006, 2012, 2018 and 2024;
2. the **overuse check**, which compares removals with NAI and flags country-years in which removals exceed NAI;
3. the **spatial component**, which spreads each national NAI over the forest of the year, weighted by the productivity of the vegetation, so that the pixels of a country add up to its national value;
4. the **exports**, which write the tables and maps to `data/outputs/` for download.

Key terms

The terms below are used throughout the page.

Term	Meaning in this document
Ecosystem service	A contribution of ecosystems to human activity. Here it is the growth of trees that can be harvested as timber.
Net annual increment (NAI)	The average annual volume growth of living trees, minus the volume of trees that die naturally. It is the quantity of new wood the forest adds in a year, and it is the accounting measure of the service.
Over bark	Volumes include the bark of the tree, as in the Eurostat questionnaire.

Term	Meaning in this document
Thousand m3	The unit of the volumes (1,000 cubic metres), as reported by Eurostat.
Forest available for wood supply (FAWS)	Forest where no environmental, social or economic restriction significantly limits wood supply. Protected areas, steep slopes or remote forests are typical restrictions.
Forest not available for wood supply (FNAWS)	Forest with such restrictions.
Corine Land Cover (CLC) accounting layers	European land cover maps at 100 m for 2000, 2006, 2012, 2018 and 2024. Forest is taken from them.
Gross Dry Matter Productivity (GDMP)	The rate at which vegetation produces dry matter, in kilograms per hectare and day, estimated from satellite images every ten days.
Removals	The volume of wood actually taken out of the forest in a year, which is close to the volume harvested.
Overuse	Removals greater than NAI. Harvesting more than the forest grows depletes the stock of timber, so the service is not sustained.
Spatial disaggregation	Splitting a national statistic over a map, using a proxy that says where the quantity is likely to occur.
NUTS	The European classification of territorial units. NUTS 0 is the country level.

Eurostat reports NAI and removals by country, not by place. The routine therefore works in two stages. It first completes the national series, which is a statistical problem, and it then places the national values on a map, which is a spatial problem.

Where the data come from

The Eurostat tables are downloaded with the `eurostat` package, which reads the same data as the Eurostat website. The download script (`scripts/03_download_eurostat.R`) keeps only the volumes in thousand cubic metres, for the years 2000 to 2024.

Two reports are consulted for the method

INCA Part III (La Notte et al., 2021) defines the service and the NAI proxy, and INCA Part II (Vallecillo et al., 2019) (chapter 4) describes how national flows are spatialised. The Eurostat ecosystem accounts handbook (`KS-01-26-019-EN-N`) (Eurostat, 2026a) complements them. Its wood chapter treats FAWS as a cultivated context measured by net increment and forest not available for wood supply (FNAWS) as an uncultivated context measured by removals. None of the references gives a formula for timber overuse, so the check below defines overuse as removals greater than NAI.

Paths are relative to the project root

`_quarto.yml` sets `execute-dir: project`, so every path below resolves from the repository root regardless of where the `.qmd` file sits. There is no need for `here::here()` or `setwd()`.

Preliminaries

Packages and helper functions

The helper functions live in `scripts/helpers_wood.R`. They keep this page short by hiding the loops and joins that are not part of the argument.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(sf)
library(terra)
library(flextable)

source("scripts/helpers_wood.R", encoding = "UTF-8")
source("scripts/helpers_results.R", encoding = "UTF-8")
```

Parameters

The reporting years are 2000, 2006, 2012, 2018 and 2024. The series is built for every year from 2000 so that the gap-filling rules can look at neighbouring years.

```r
years_all <- 2000:2024
years_report <- c(2000, 2006, 2012, 2018, 2024)

# Inputs (committed to the repository)
eurostat_file <- "data/inputs/eurostat/for_vol_efa_thsm3.rds"
countries_file <- "data/inputs/gis/NUTS_RG_01M_2024_3035_LEVL0.gpkg"
faws_area_file <- "data/inputs/faws/faws_area_by_country.rds"
weights_dir <- "data/inputs/weights"
nuts_weights_file <- "data/inputs/weights/weight_by_nuts.rds"

# Outputs
results_dir <- "data/outputs/results"
gis_dir <- "data/outputs/gis"
dir.create(results_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(gis_dir, showWarnings = FALSE, recursive = TRUE)
```

Preparing the inputs

Four inputs are too heavy or too raw to live in the repository as downloaded. Each one is reduced by a short script in `scripts/`, which is run once and whose compact product is committed. The scripts read from `refs/`, which is not versioned, so anyone who wants to repeat the reduction needs to place the original files there.

Script	Source	Product in `data/inputs/`
`01_prepare_nuts.R`	Eurostat GISCO NUTS 2024, 1:1 million, shipped with the INCA plugin (Eurostat GISCO, 2024)	`gis/NUTS_RG_01M_2024_3035_LEVL0.gpkg`, country polygons in EPSG:3035
`02_prepare_faws.R`	FAWS 2020 map, 100 m, EU27 (European Commission, Joint Research Centre, 2026)	`faws/faws_area_by_country.rds` (exact hectares of FAWS and FNAWS by country at 100 m)
`03_download_eurostat.R`	Eurostat `for_vol_efa`, thousand m3, NAI and removals (Eurostat, 2026b)	`eurostat/for_vol_efa_thsm3.rds` and `.csv`

Script	Source	Product in `data/inputs/`
`05_prepare_weights.R`	Corine Land Cover accounting layers, 100 m, version 21 (European Environment Agency, 2026), and annual Gross Dry Matter Productivity of the Copernicus Land Monitoring Service, 1 km and 300 m, version 2 (Copernicus Land Monitoring Service, n.d.-a, n.d.-b)	`weights/clc_gdmp_weight_1km_<year>.tif`, the spatial key of each reporting year at 1 km (about 12 MB each)
`06_prepare_nuts_weights.R`	The weights above and the NUTS 2024 regions	`weights/weight_by_nuts.rds`, the share of every NUTS2 and NUTS3 region in the weight of its country, by year

The FAWS map has two values, 1 and 2. The coding (1 for FAWS, 2 for forest not available) is inferred from the pixel counts and the accompanying report (Avitabile et al., 2023), since the files do not state it. The FAWS map is used only to measure the share of forest that is available in each country, for the sensitivity check below. The annual productivity is the sum of the 36 ten-day values of the year, each multiplied by its number of days. Script `05_prepare_weights.R` also keeps the 100 m weights in `refs/weights_100m/`, which are used for the 100 m maps and are not versioned.

```
source("scripts/01_prepare_nuts.R", encoding = "UTF-8")
source("scripts/02_prepare_faws.R", encoding = "UTF-8")
source("scripts/03_download_eurostat.R", encoding = "UTF-8")
source("scripts/05_prepare_weights.R", encoding = "UTF-8")
source("scripts/06_prepare_nuts_weights.R", encoding = "UTF-8")
```

Map boundaries

Country maps use the NUTS boundaries of Eurostat GISCO. © EuroGeographics for the administrative boundaries.

Statistical component

Loading Eurostat data

The data come from the European Forest Accounts questionnaire, which member states answer every year. They are partial in practice, because many countries cannot split NAI between available and unavailable forest, especially in the early years. The extract has one row per country, year, flow (`NAI` or `RMOV`) and forest class. `FAWS` is forest available for wood supply, `FNAWS` forest not available, and `FOR` total forest. Only the 27 member states are kept, and the EU aggregate published by Eurostat is held aside to check the totals later.

```
for_vol_efa <- readRDS(eurostat_file)

countries <- st_read(countries_file, quiet = TRUE)
eu27 <- countries$geo[countries$eu27]

eu_aggregate <- for_vol_efa |>
  filter(geo == "EU27_2020")

for_vol_efa <- for_vol_efa |>
  filter(geo %in% eu27)
```

A small function picks one series of the extract. It keeps the later steps to a single line.

```
pick_series <- function(flow_name, class_name) {
  for_vol_efa |>
    filter(flow == flow_name, forest_class == class_name)
}
```

How much was reported

The table counts the member states that reported each series in the five reporting years. NAI of FAWS is reported by very few countries before 2018, whereas NAI of total forest and removals cover nearly all of them. This is the reason for the gap-filling rule in the next section.

```
for_vol_efa |>
  filter(
    year %in% years_report,
    forest_class %in% c("FAWS", "FOR")
  ) |>
  count(flow, forest_class, year) |>
```

```
  pivot_wider(names_from = year, values_from = n, values_fill = 0) |>
  rename(Flow = flow, `Forest class` = forest_class) |>
  show_table(digits = 0)
```

Flow	Forest class	2006	2012	2018	2024	2000
NAI	FAWS	2	6	14	2	0
NAI	FOR	26	26	26	2	26
RMOV	FAWS	26	26	26	2	26
RMOV	FOR	26	26	26	2	26

Gap filling NAI of FAWS

The INCA plugin fills the NAI of FAWS in the order below, and the routine does the same so that results are comparable with the plugin. The helper `fill_series()` applies the rules and labels each value with its origin.

1. Use the NAI of FAWS where the country reported it (`reported`).
2. Otherwise use the NAI of total forest (`fallback`). Total forest includes the forest that is not available, so this is an upper bound for the flow of FAWS, but it is the closest reported figure.
3. Remaining gaps take the mean of the nearest earlier and later value of the same country (`interpolated`). Forest growth changes slowly, so a value between two reported years is a reasonable estimate. Gaps at the start or end of a series repeat the nearest value (`carried`), which assumes no change beyond the data.

```
nai_filled <- fill_series(
  primary = pick_series("NAI", "FAWS"),
  fallback = pick_series("NAI", "FOR"),
  geos = eu27,
  years = years_all
)
```

The table shows how many of the 27 values of each reporting year come from each rule. The 2024 row is dominated by carried values, because Eurostat holds 2024 for very few countries.

```r
nai_filled |>
  filter(year %in% years_report) |>
  count(year, source) |>
  pivot_wider(names_from = source, values_from = n, values_fill = 0) |>
  rename(Year = year) |>
  show_table(digits = 0)
```

Year	fallback	carried	reported
2000	26	1	0
2006	24	1	2
2012	20	1	6
2018	12	1	14
2024	0	25	2

Check against the Eurostat aggregate

The sum of the 27 country values is compared with the EU27 aggregate that Eurostat publishes for the same indicator. The aggregate for FAWS is Eurostat's own estimate and differs from the sum of the reported country values, so some differences are expected. The table is a plausibility check, and values are in million cubic metres.

```r
eu_check <- nai_filled |>
  group_by(year) |>
  summarise(`Sum of countries` = sum(value, na.rm = TRUE) / 1000) |>
  left_join(
    eu_aggregate |>
      filter(flow == "NAI", forest_class %in% c("FAWS", "FOR")) |>
      mutate(
        series = paste("Eurostat EU27", forest_class),
        million_m3 = thousand_m3 / 1000
      ) |>
      select(year, series, million_m3) |>
      pivot_wider(names_from = series, values_from = million_m3),
    by = "year"
  ) |>
  filter(year %in% years_report) |>
  rename(Year = year)
```

```r
show_table(eu_check, digits = 0)
```

Year	Sum of countries	Eurostat EU27 FAWS	Eurostat EU27 FOR
2000	912	891	912
2006	900	838	900
2012	855	814	871
2018	805	784	849
2024	736		

A sensitivity check on the total forest fallback

NAI of total forest includes the forest not available for wood supply, so using it in place of the NAI of FAWS probably overstates the flow in the countries that need the fallback. The FAWS map gives the share of forest that is available in 2020 for every country. As a sensitivity check, the fallback values are multiplied by that share. The adjusted series is not used in the main results, and it is exported in the summary workbook for comparison.

```r
faws_area <- readRDS(faws_area_file) |>
  mutate(faws_share = faws_ha / (faws_ha + fnaws_ha))

nai_adjusted <- nai_filled |>
  left_join(faws_area |> select(geo, faws_share), by = "geo") |>
  mutate(
    value_adjusted = if_else(source == "fallback", value * faws_share, value)
  )

nai_adjusted |>
  filter(year %in% years_report) |>
  group_by(year) |>
  summarise(
    `INCA rule` = sum(value) / 1000,
    `Fallback scaled by FAWS share` = sum(value_adjusted) / 1000
  ) |>
  mutate(`Difference (%)` = 100 * (`Fallback scaled by FAWS share` / `INCA rule` - 1)) |>
  rename(Year = year) |>
  show_table(digits = 1)
```

Year	INCA rule	Fallback scaled by FAWS share	Difference (%)
2000	912.3	819.4	-10.2
2006	899.5	811.4	-9.8
2012	855.0	776.3	-9.2
2018	805.1	754.1	-6.3
2024	735.7	735.7	0.0

Overuse check

NAI is the wood that the forest grows in a year, and removals are the wood taken out. When removals stay above NAI the standing stock shrinks, which means that the service is being used faster than it renews. Overuse is reported when removals exceed NAI. Removals are used and not fellings, because fellings are not part of `for_vol_efa`. The routine takes the removals from forest available for wood supply (`RMOV` of `FAWS`), with the same fallback and gap-filling rules as NAI, so both sides of the comparison cover the same forest.

The comparison is made year by year. A single year above 1 can be a temporary peak, for example after a storm, and a sustained ratio above 1 is the stronger signal. The ratio of removals to NAI is kept in the table, in addition to the flag, for that reason.

```
removals_filled <- fill_series(
  primary = pick_series("RMOV", "FAWS"),
  fallback = pick_series("RMOV", "FOR"),
  geos = eu27,
  years = years_all
)

overuse <- check_overuse(nai_filled, removals_filled)
```

The next table lists the country-years with overuse among the reporting years. When the NAI or the removals of a country-year were carried forward or taken from the fallback, the flag deserves a second look, and the source columns show it.

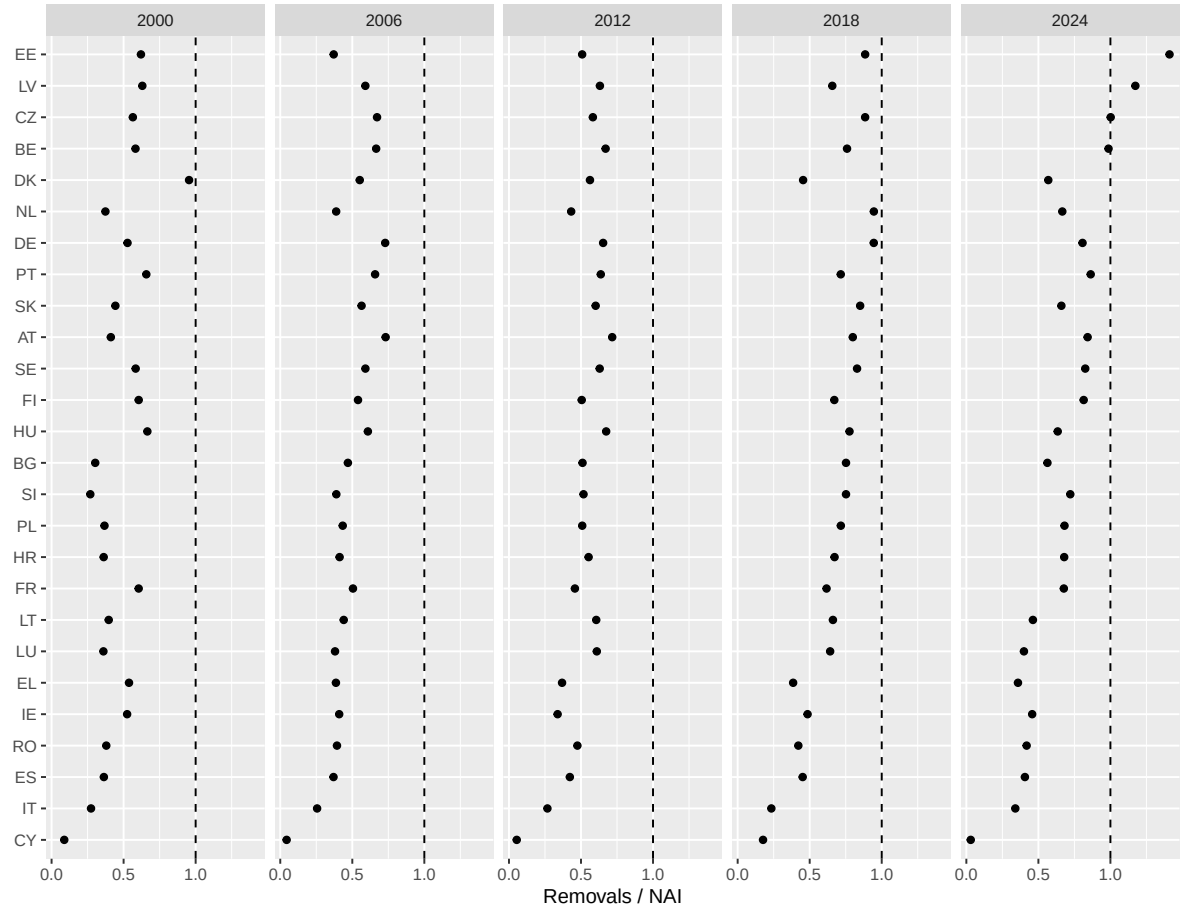
```
overuse |>
  filter(year %in% years_report, overuse) |>
  transmute(
    Year = year,
    Country = geo,
```

```
    `NAI (1000 m3)` = nai,
    `Removals (1000 m3)` = removals,
    `Removals / NAI` = removals_to_nai,
    `NAI source` = as.character(nai_source),
    `Removals source` = as.character(removals_source)
  ) |>
  arrange(Year, desc(`Removals / NAI`)) |>
  show_table(digits = 2)
```

Year	Country	NAI (1000 m3)	Removals (1000 m3)	Removals / NAI	NAI source	Removals source
2024	EE	8,200.00	11,564.00	1.41	carried	carried
2024	LV	16,126.60	18,920.00	1.17	carried	carried
2024	CZ	20,458.35	20,495.73	1.00	carried	carried

The chart shows the ratio of removals to NAI for the countries in the reporting years. Values to the right of the dashed line are overuse. Malta has only zero values in the extract, so its ratio is undefined and it is left out of the chart.

```
overuse |>
  filter(year %in% years_report, is.finite(removals_to_nai)) |>
  mutate(geo = reorder(geo, removals_to_nai, FUN = max)) |>
  ggplot(aes(x = removals_to_nai, y = geo)) +
  geom_vline(xintercept = 1, linetype = "dashed") +
  geom_point() +
  facet_wrap(~year, nrow = 1) +
  labs(x = "Removals / NAI", y = NULL)
```



Spatial component

The spatial key

The spatial key follows the method of the INCA tool and of INCA Part II. Chapter 4 of INCA Part II spreads national flows over the forest classes of Corine Land Cover with dry matter productivity as a proxy (Vallecillo et al., 2019). The INCA wood provision tool does the same, multiplying a forest mask by an annual productivity layer and spreading each national value in proportion to the result. The Earth observation roadmap of the INCA project names the Gross Dry Matter Productivity (GDMP) of the Copernicus Land Monitoring Service as the proxy for the provisioning services (Zurbarán Nucci et al., 2025). A decision was made to follow this method. Cells with more productive vegetation add more wood, so they receive a larger part of the national NAI.

The weight of a 100 m cell is 1 if it is forest in the Corine accounting layer of the year (broad-leaved, coniferous and mixed forest and transitional woodland and shrub, CLC 311, 312, 313 and 324) and 0 otherwise, times the annual productivity of the same year. The sources of productivity differ by year. The 1 km product, version 2, covers 2000, 2006 and 2012 (Copernicus Land Monitoring Service, n.d.-a). The 300 m product, version 2, covers 2018 and 2024 (Copernicus Land Monitoring Service, n.d.-b). The weights only share a national value among the cells of a country, so a difference in level between the two products does not change any national total. The weight of a 1 km cell is the sum of the weights of the 100 cells of 100 m inside it.

Inside each country, the national NAI is spread in proportion to the weight of every cell. Cells without forest get no flow, and the cells of a country add up to its national value. The forest used for the weight is all forest and not only the forest available for wood supply. INCA Part II and the INCA tool do not restrict the spatial key to FAWS, because the thresholds that define availability are not common across countries and a map built on them can mislead (Vallecillo et al., 2019). The statistic that is spread is still the NAI of FAWS.

Three choices in this step deserve a short explanation. First, the maps use the projection EPSG:3035 (ETRS89 Lambert azimuthal equal-area), in which every cell covers the same area on the ground, so a share of a cell and a count of cells translate directly into areas. Second, the weights are built at 100 m and summed to 1 km, which keeps the files in the repository small. The maps at 100 m are kept outside the repository because of their size. Third, the sum of each country is conserved, which means that mapping redistributes the national NAI and neither adds nor removes any.

```
eu_countries <- vect(countries_file)
eu_countries <- eu_countries[eu_countries$eu27, ]
weights <- lapply(years_report, \(y) rast(file.path(weights_dir, sprintf("clc_gdmp_weight_1k
names(weights) <- years_report
```

Country grid and weights

Each 1 km cell is assigned to a country, and the weights of every country are added once per year.

```
country_ids <- make_country_ids(eu_countries, weights[[1]])
weight_sums <- lapply(weights, country_weight_sums, country_ids = country_ids)
```

Spreading the national NAI

For each reporting year, the helper `spread_totals()` multiplies every cell by its country factor (national NAI divided by the weight total of the country). The result is in thousand cubic metres

per 1 km cell, and each year is saved as a compressed GeoTIFF in `data/outputs/gis/`.

```
for (yr in years_report) {
  flow_map <- spread_totals(
    totals = filter(nai_filled, year == yr),
    weight = weights[[as.character(yr)]],
    country_ids = country_ids,
    weight_sums = weight_sums[[as.character(yr)]]
  )
  names(flow_map) <- paste0("nai_clc_gdmp_", yr)
  writeRaster(
    flow_map,
    file.path(gis_dir, sprintf("wood_nai_clc_gdmp_1km_%s.tif", yr)),
    overwrite = TRUE,
    datatype = "FLT4S",
    gdal = c("COMPRESS=DEFLATE", "PREDICTOR=3", "TILED=YES")
  )
}
```

Conservation check

The sum of each map must equal the sum of the national values of that year. The check reads the maps back from disk, so it also verifies the files that are delivered.

```
nai_maps <- rast(file.path(gis_dir, sprintf("wood_nai_clc_gdmp_1km_%s.tif", years_report)))

national_totals <- nai_filled |>
  filter(year %in% years_report) |>
  group_by(year) |>
  summarise(total = sum(value) / 1000) |>
  pull(total)

tibble(
  Year = years_report,
  `Sum of maps (million m3)` = global(nai_maps, "sum", na.rm = TRUE)[[1]] / 1000,
  `Sum of national values (million m3)` = national_totals
) |>
  mutate(
    `Difference (%)` = 100 * (`Sum of maps (million m3)` / `Sum of national values (million m
  ) |>
  show_table(digits = 2)
```

Year	Sum of maps (million m3)	Sum of national values (million m3)	Difference (%)
2,000.00	912.29	912.29	-0.00
2,006.00	899.54	899.54	0.00
2,012.00	854.97	854.97	0.00
2,018.00	805.08	805.08	0.00
2,024.00	735.73	735.73	0.00

A second check lists the countries that have a national value and no forest weight in some year. Such a country would lose its flow. A national NAI of zero, as in Malta, is not a loss.

```
lapply(names(weight_sums), \(y) mutate(weight_sums[[y]], year = as.integer(y))) |>
  bind_rows() |>
  right_join(nai_filled |> filter(year %in% years_report), by = c("geo", "year")) |>
  filter((is.na(weight_sum) | weight_sum <= 0) & value > 0) |>
  select(Year = year, Country = geo, `NAI (1000 m3)` = value) |>
  show_table(digits = 0)
```

Year	Country	NAI (1000 m3)

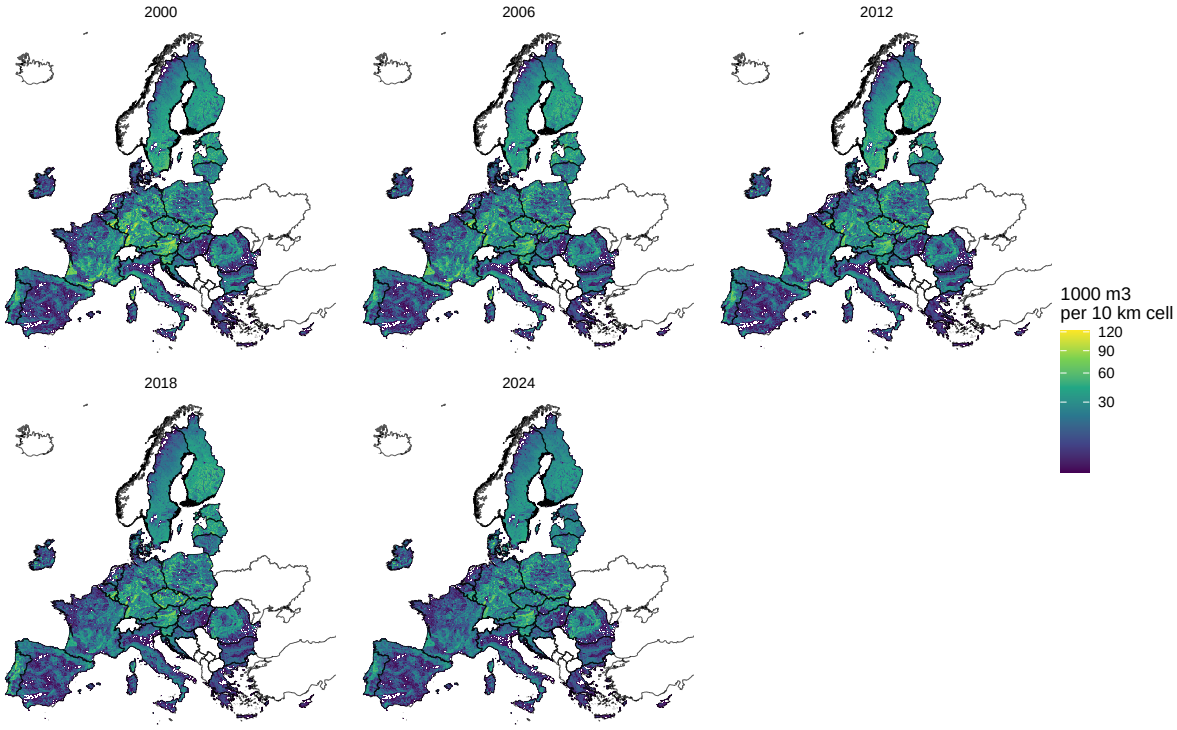
Maps

The maps are shown at 10 km (sum of the 100 cells of 1 km) so that they draw quickly. The delivered files keep the 1 km resolution, and the 100 m maps are available in the external folder listed in the downloads.

```
map_data <- aggregate(nai_maps, fact = 10, fun = "sum", na.rm = TRUE) |>
  as.data.frame(xy = TRUE) |>
  pivot_longer(starts_with("nai_clc_gdmp_"), names_to = "year", values_to = "nai") |>
  mutate(year = sub("nai_clc_gdmp_", "", year)) |>
  filter(nai > 0)

ggplot(map_data) +
  geom_tile(aes(x = x, y = y, fill = nai)) +
  geom_sf(data = countries, fill = NA, linewidth = 0.1) +
  scale_fill_viridis_c(transform = "sqrt", name = "1000 m3\nper 10 km cell") +
  facet_wrap(~year, ncol = 3) +
  coord_sf(xlim = c(2.6e6, 6.6e6), ylim = c(1.4e6, 5.4e6), datum = NA) +
```

```
  theme_void() +
  labs(x = NULL, y = NULL)
```



Maps at 100 m

The maps at 100 m use the same key without the summation to 1 km, so a forest cell of 100 m receives its share of the national NAI. They are written by `scripts/07_write_maps_100m.R`, with the same conservation check, and they are too large for the repository. Each file weighs about 0.6 GB because continuous values compress poorly. They are delivered in an external folder together with a README that gives the grid, units, method and totals.

Exports

The tables are saved as `.rds`, `.csv` and `.xlsx` through `export_table()`, and a workbook with the reporting-year tables is saved for users. A GeoPackage of the countries carries the reporting-year values for use in GIS.

```
nai_export <- nai_filled |>
  rename(nai_faws_1000m3 = value, nai_source = source)

export_table(nai_export, "wood_nai_faws_country_year")

export_table(
  overuse |> mutate(across(c(nai_source, removals_source), as.character)),
  "wood_overuse_country_year"
)
```

```
summary_sheets <- list(
  nai_report_years = nai_export |>
    filter(year %in% years_report) |>
    select(geo, year, nai_faws_1000m3) |>
    pivot_wider(names_from = year, values_from = nai_faws_1000m3),
  overuse_report_years = overuse |>
    filter(year %in% years_report) |>
    mutate(across(c(nai_source, removals_source), as.character)),
  sensitivity_fallback_scaled = nai_adjusted |>
    filter(year %in% years_report) |>
    transmute(
      geo, year,
      inca_rule = value,
      scaled_by_faws_share = value_adjusted,
      source = as.character(source)
    ),
  faws_area_by_country = faws_area
)

writexl::write_xlsx(summary_sheets, file.path(results_dir, "wood_provision_summary.xlsx"))
```

The results workbook has the tables of the actual flow and of the overuse for NUTS0, NUTS2 and NUTS3. Regions share the national value in proportion to their weight in the spatial key of the year, so they add up to the country. The sheets in monetary terms are added by the monetary valuation document.

```
write_results_workbook(
  overuse, readRDS(nuts_weights_file), years_report, base_year = 2010,
  path = file.path(results_dir, "wood_provision_results.xlsx")
)
```

```
country_values <- overuse |>
  filter(year %in% years_report) |>
  select(geo, year, nai, removals, overuse) |>
  pivot_wider(
    names_from = year,
    values_from = c(nai, removals, overuse),
    names_glue = "{.value}_{year}"
  )

countries |>
  filter(eu27) |>
  left_join(country_values, by = "geo") |>
  st_write(
    file.path(gis_dir, "wood_provision_countries.gpkg"),
    layer = "countries",
    delete_dsn = TRUE,
    quiet = TRUE
  )
```

Downloads

`_quarto.yml` lists `data/outputs/` under `project.resources`, so these files are copied into the published site. The links below are relative to this page.

```
tables_dl <- tibble(
  File = c(
    "wood_nai_faws_country_year.xlsx",
    "wood_nai_faws_country_year.rds",
    "wood_overuse_country_year.xlsx",
    "wood_overuse_country_year.rds",
    "wood_provision_summary.xlsx",
    "wood_provision_results.xlsx"
  ),
  Folder = "results",
  Description = c(
```

```
    "National NAI of FAWS by year, 2000 to 2024, with the origin of every value",
    "The same panel as an R object",
    "NAI, removals, balance, ratio and overuse flag by country and year",
    "The same panel as an R object",
    "Reporting-year tables and the fallback sensitivity check, one sheet each",
    "Results tables for NUTS0, NUTS2 and NUTS3 in physical terms, with the monetary sheets ac
  )
)

maps_dl <- tibble(
  File = c(
    sprintf("wood_nai_clc_gdmp_1km_%s.tif", years_report),
    "wood_provision_countries.gpkg",
    "100 m maps: flow and value, 2000 to 2024 (Dropbox folder)"
  ),
  Folder = c(rep("gis", length(years_report) + 1), "external"),
  Description = c(
    sprintf("NAI in %s, thousand m3 per 1 km cell, EPSG:3035", years_report),
    "EU27 country polygons with NAI, removals and overuse flag for the reporting years",
    "Maps of the NAI and of its value at 100 m for the five reporting years. They are too lar
  )
)

downloads <- bind_rows(tables_dl, maps_dl) |>
  mutate(Path = if_else(
    Folder == "external",
    "https://www.dropbox.com/scl/fo/v4hxro4cb600n8jj98j28/AIOkKqQsECgwZdttaoXRx3s?rlkey=lcc6a
    file.path("../../../data/outputs", Folder, File)
  ))

flextable(downloads, col_keys = c("File", "Description")) |>
  compose(
    j = "File",
    value = as_paragraph(hyperlink_text(x = File, url = Path))
  ) |>
  fontsize(size = 9, part = "all") |>
  autofit()
```

File	Description
wood_nai_faws_country_year.xlsx	National NAI of FAWS by year, 2000 to 2024, with the origin of every value

File	Description
wood_nai_faws_country_year.rds	The same panel as an R object
wood_overuse_country_year.xlsx	NAI, removals, balance, ratio and overuse flag by country and year
wood_overuse_country_year.rds	The same panel as an R object
wood_provision_summary.xlsx	Reporting-year tables and the fallback sensitivity check, one sheet each
wood_provision_results.xlsx	Results tables for NUTS0, NUTS2 and NUTS3 in physical terms, with the mone
wood_nai_clc_gdmp_1km_2000.tif	NAI in 2000, thousand m3 per 1 km cell, EPSG:3035
wood_nai_clc_gdmp_1km_2006.tif	NAI in 2006, thousand m3 per 1 km cell, EPSG:3035
wood_nai_clc_gdmp_1km_2012.tif	NAI in 2012, thousand m3 per 1 km cell, EPSG:3035
wood_nai_clc_gdmp_1km_2018.tif	NAI in 2018, thousand m3 per 1 km cell, EPSG:3035
wood_nai_clc_gdmp_1km_2024.tif	NAI in 2024, thousand m3 per 1 km cell, EPSG:3035
wood_provision_countries.gpkg	EU27 country polygons with NAI, removals and overuse flag for the reporting ye
100 m maps: flow and value, 2000 to 2024 (Dropbox folder)	Maps of the NAI and of its value at 100 m for the five reporting years. They are t

Limitations and proposals to refine the spatial key

- The key combines where forest is with the productivity of the vegetation. Productivity measures the growth of vegetation in general and does not distinguish stands that are available for wood supply from those that are not, so some flow is placed on forest that is not harvested.
- The maps of 2000, 2006 and 2012 use a 1 km productivity product and those of 2018 and 2024 use a 300 m product, so the spatial detail of the two groups of years differs, and differences between years partly reflect the change of product.
- The Corine accounting layers have a minimum mapping unit of 25 ha. The class of transitional woodland and shrub is included in the forest mask, as in the INCA tool.
- NAI of FAWS is reported by few countries before 2018. The fallback to total forest probably overstates the flow, as the sensitivity check shows. The INCA plugin replaces missing values with estimates from the JRC carbon budget model, which are not available in this repository.
- The 2024 values mostly repeat the latest available year, since Eurostat holds 2024 for very few countries. They should be refreshed when the dataset is updated.
- The boundary layers are aggregated to 1 km, so cells on the coast and on borders are assigned to one country by their centre point.

Further candidate refinements include a mask of forest availability, which the INCA tool accepts

as an optional set of weights, and the biomass available for wood supply map as an additional weight. The monetary valuation that follows the physical accounts is in its own document.

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